



Diffusion Model-Based Generation of Three-Dimensional Multiphase Pore-Scale Images

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Abstract

We propose a diffusion model-based machine learning method for generating three-dimensional images of both the pore space of rocks and the fluid phases within it. This approach overcomes the limitations of current methods, which are restricted to generating only the pore space. Our reconstructed images accurately reproduce multiphase fluid pore-scale details in water-wet Bentheimer sandstone, matching experimental images in terms of two-point correlation, porosity, and fluid flow parameters. This method outperforms generative adversarial networks with a broader and more accurate parameter range. By enabling the generation of multiphase fluid pore-scale images of any size subject to computational constraints, this machine learning technique provides researchers with a powerful tool to understand fluid distribution and movement in porous materials without the need for costly experiments or complex simulations. This approach has wide-ranging potential applications, including carbon dioxide and underground hydrogen storage, the design of electrolyzers, and fuel cells.

Keywords Multiphase image generation · Diffusion models · Flow in porous media · Transformer · Curvature

1 Introduction

An in-depth understanding of flow in porous media is important for applications such as water resources, carbon dioxide (CO₂) sequestration, underground hydrogen (H₂) storage, hydrocarbon recovery, and fuel cells (Class et al. 2009; Boot-Handford et al. 2014; Hematpur et al. 2023; Li et al. 2023; Blunt et al. 2024). Currently, common non-destructive methods for observing flow in porous media include X-ray computed tomography, microfluidics, magnetic resonance imaging (MRI), scanning electron microscopy (SEM), and other visualization techniques (Blunt et al. 2013; Breugem et al. 2014; Gladden and Sederman 2017; Anbari et al. 2018; Sheng et al. 2020; Wang et al. 2024; Qin et al. 2024; Cai et al. 2024). These approaches have advanced our understanding of fluid behavior in porous structures.

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However, each method comes with trade-offs in terms of field of view and dimensionality. For example, while SEM provides very high-resolution images, it only yields two-dimensional (2D) slices, making it challenging to capture the three-dimensional (3D) connectivity that governs fluid flow (Zhu et al. 2022; Hu et al. 2023). Moreover, pore structures in rock formations are directionally heterogeneous. For instance, Xia et al. (2019) showed that pore structure parameters vary significantly with direction, exerting a strong influence on fluid transport. Such complexity cannot be fully represented by examining a few 2D sections alone, underscoring the need for truly 3D data. At the same time, no single imaging technique currently provides both a sufficiently large field of view and high resolution (Zhu et al. 2024). Compounding these challenges, multiphase flow experiments incorporating imaging often span several months and risk damaging samples (Goodarzi et al. 2024). Together, these factors highlight the importance of developing reliable methods to reconstruct 3D images-whether from 2D slices or smaller 3D subvolumes,thereby improving our understanding of subsurface flow processes.

Before the advent of machine learning, conventional reconstruction methods included process-based simulation, statistical simulation, stochastic reconstruction, and multiple-point statistics (Okabe and Blunt 2004, 2005; Yang et al. 2024; Xiao et al. 2024). Although these methods laid an important foundation, they often suffered from slow reconstruction speeds, limited accuracy, and difficulty handling highly heterogeneous samples. With the establishment of digital rock data repositories (e.g., the Digital rocks portal) and the increased availability of multiphase flow imaging data, research integrating machine learning and data-driven approaches have accelerated. Generative adversarial networks (GANs) have been employed successfully to produce pore-space images (Li et al. 2023). Since the pioneering work by Mosser et al. (2017), which demonstrated the feasibility of using GANs to reconstruct relatively simple sandstone and carbonate structures, machine learning-based digital rock reconstructions have experienced explosive growth. Subsequent studies have largely relied on GAN architectures, which can be grouped into two main strategies: 2D-to-3D and 3D-to-3D generation. For instance, You et al. (2021) used a progressive growing GAN to generate individual 2D slices, later stacked into a 3D volume. Valsecchi et al. (2020) combined a 2D discriminator with a 3D generator to produce reliable 3D images even when only 2D data were available. Volkhonskiy et al. (2019) integrated an autoencoder with a GAN to improve the quality of reconstructed 3D images from 2D slices, while Shams et al. (2021) adopted an ST-CGAN architecture to better capture heterogeneity. Further refinements, including techniques including Fréchet inception distance (FID) and SliceGAN, have improved the reliability of 3D reconstructions from limited 2D input (Amiri et al. 2024). Nevertheless, when sufficient 3D training data are available, direct 3D-to-3D methods typically yield more reliable reconstructions. Examples include Zhao et al. (2021), who used a DCGAN with both 3D generators and discriminators, and Zhang and Ni (2023), who employed concurrent GANs and residual blocks for 3D generation. More recently, Zhu et al. (2024) introduced the improved pyramid Wasserstein GAN (IPWGAN) model, achieving high-resolution, centimeter-scale rock reconstructions at micrometer resolution for both complex sandstones and carbonates.

Two-point correlation functions and linear path functions are commonly used to characterize voxel-level spatial connectivity (Haslach 2002; Torquato and Yeong 1998), while parameters derived from Minkowski functionals such as porosity, specific surface area, curvature, and Euler characteristic-help evaluate the flow-related properties of the reconstructed porous media (Vogel et al. 2010; Armstrong et al. 2019; Sadeghnejad et al. 2023). Following this established practice, we use these metrics in our work to directly assess the realism of reconstructed samples in terms of their fluid transport characteristics. The

generation of synthetic pore spaces has enabled the production of large sets of images that simultaneously achieve high resolution and large field of view (Zhu et al. 2019; Shams et al. 2021; Yang et al. 2022). Beyond structural reconstructions, machine learning has also been employed to predict properties including permeability, improve image segmentation, and enhance resolution (Da Wang et al. 2021; Zhang et al. 2022).

Diffusion models have emerged as powerful alternatives in the generative modeling landscape (Sohl-Dickstein et al. 2015; Ho et al. 2020). Unlike GANs, which can suffer from training instability and mode collapse, diffusion models gradually transform random noise into high-fidelity images through a controlled iterative denoising process. This approach tends to yield greater stability, a higher level of diversity, and a more faithful representation of complex structures and spatial correlations. These advantages have led diffusion models to achieve state-of-the-art performance in a range of image processing tasks, including image synthesis and inpainting.

Building upon these strengths, our study aims to move beyond single-phase reconstructions toward generating fully 3D multiphase pore-scale images. By accounting for multiple fluid phases, we more accurately capture fluid–fluid and fluid–solid interactions, thus moving toward an artificial intelligence (AI)-driven paradigm for investigating multiphase flow in porous media. This capability enables us to explore a potentially wider range of conditions such as varying wettability states (Lv et al. 2024), flow regimes, and fluid properties—than conventional methods or datasets typically allow.

The remainder of this paper is organized as follows: Section 2 details the methodology and dataset used in this study. Section 3 analyzes the results from multiple perspectives, verifying their fidelity and complexity. Section 4 discusses potential improvements, limitations, and future directions. Finally, the last section summarizes our key findings and their implications for research on multiphase flow in porous media.

2 Methodology and Datasets

Diffusion models are an advanced category of generative machine learning methods that, unlike generative adversarial networks (GANs), do not require a discriminator for training (Aggarwal et al. 2021; Croitoru et al. 2023). In contrast to GANs—which rely on a delicate balance between generator and discriminator, often resulting in training instability and sensitivity to hyperparameters, diffusion models optimize a direct objective, thus simplifying the training process and improving robustness. Essentially, diffusion models iteratively corrupt clean data by adding noise and then learn to remove it step-by-step, ultimately producing highly detailed outputs ranging from realistic images to coherent text sequences.

Their fundamental mechanism involves a forward diffusion and a backward denoising process. In the forward stage, Gaussian noise is progressively added over multiple steps, transforming a clean image into an almost entirely noise-contaminated version (Fig. 1a). Formally, at each forward step t :

$$x_t = \sqrt{\alpha_t} \times x_{t-1} + \sqrt{1 - \alpha_t} \times \varepsilon_t, \quad (1)$$

where x_t is the corrupted image at step t , x_{t-1} is image at the previous step, ε_t is Gaussian noise, and α_t is a noise-weighting coefficient that grows with t .

The backward denoising stage, as shown in Fig. 1b–c, reverses this process. Starting from a highly corrupted image, the model predicts and removes noise at each step,

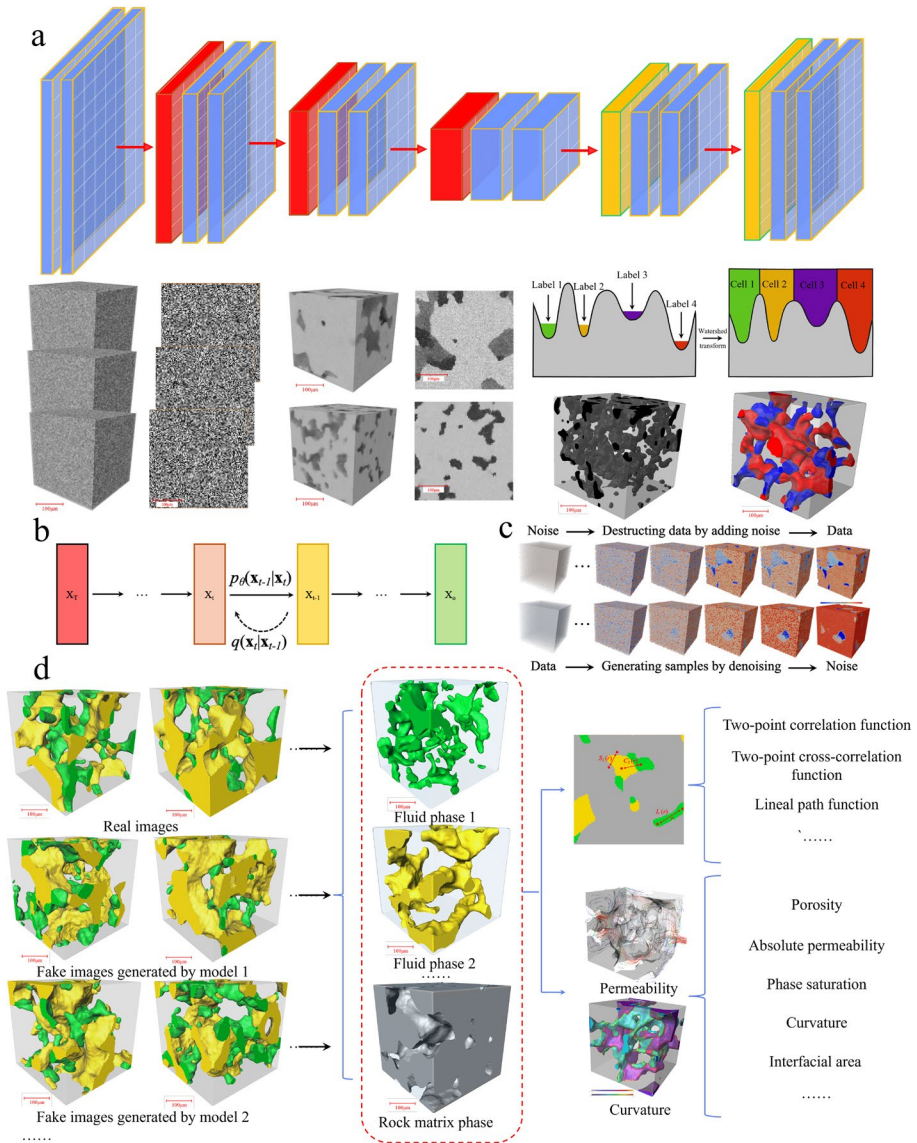


Fig. 1 Overview of the diffusion model-based generation of multiphase fluid pore-scale images. **a** The complete training process using a U-Net architecture: blue blocks represent input images and extracted features; red blocks are encoder stages (downsampling and feature transformation); and yellow blocks are decoder stages (upsampling and feature fusion). Outputs include 3D and 2D unsegmented images, which can then be segmented into different fluid phases using methods such as the watershed algorithm. **b** Conceptual depiction of the forward diffusion ($q(x_t|x_{t-1})$) and reverse denoising ($p_\theta(x_{t-1}|x_t)$) processes. **c** The top row shows progressive denoising during sample generation, while the bottom row shows the reverse direction representing the training process. This illustrates how the model learns to transform noisy data into clean pore-scale images. **d** After generating synthetic images, they are segmented and compared with real pore-scale images. A variety of geometric and flow-related metrics are evaluated to assess fidelity

gradually recovering the original image. This involves training a neural network to estimate the noise $\varepsilon_\theta(x_t, t)$. The backward update rule is:

$$x_{t-1} = \sqrt{\alpha_{t-1}} \left(x_t - \frac{\sqrt{1-\alpha_t}}{\sqrt{1-\alpha_{t-1}}} \varepsilon_\theta(x_t, t) \right) + \sqrt{1-\alpha_{t-1}} \varepsilon_\theta(x_t, t). \quad (2)$$

The network is trained using the loss function:

$$L = \mathbb{E}_{x_0, \varepsilon} \left[|\varepsilon - \varepsilon_\theta(\sqrt{\alpha_t} x_0 + \sqrt{1-\alpha_t} \varepsilon, t)|^2 \right], \quad (3)$$

where x_0 is the original clean image and ε is sampled Gaussian noise. Minimizing this L_2 loss makes the model accurately predict the noise at each step. Although the L_2 norm does not fully capture higher-order spatial structures, as we show later this ensures stable training and the generation of representative images.

DDIM (Denoising diffusion implicit model) (Song et al. 2021) is a diffusion model variant designed to improve sampling efficiency and speed. As illustrated in Fig. 1d, DDIM introduces a latent variable into the backward process, enabling more flexible and efficient image generation. In each denoising step, the model estimates the noise and updates a latent variable z_t , balancing the original clean signal and estimated noise:

$$z_t = \sqrt{\alpha_t} \times z_0 + \sqrt{1-\alpha_t} \times \varepsilon_\theta(x_t, t), \quad (4)$$

where z_0 is the initial latent representation. By adjusting α_t , the model can efficiently sample from the learned distribution.

During the reverse denoising process, architectures such as a U-shaped convolutional neural network (U-Net) or transformers are commonly employed to estimate noise (Siddique et al. 2021; Han et al. 2022). U-Nets excel at extracting local features, while transformers capture global dependencies—crucial for generating multiphase images where fluid distributions in different pores influence each other. In this study, we integrate multiple 3D transformer layers at specific network levels (Han et al. 2022) to balance global feature extraction and memory usage. Figure 1 overviews the entire modeling, generation, and evaluation workflow for multiphase fluid images using diffusion models.

We validate our approach using a widely recognized benchmark dataset of water-wet Bentheimer sandstone (Lin et al. 2018). This dataset includes images acquired after successive fluid injection steps: the pore space fully saturation with gas fully saturated with brine (wetting phase 1) at steady state during both brine and oil (phase 2) injection at different fractional flows f_w . We focus on images corresponding to $f_w = 0.15$. The voxel size is 3.58 μm , and we analyze a $1000 \times 1000 \times 3000$ voxel subvolume. After applying the watershed algorithm in Avizo software, we segment each fluid phase from the solid matrix.

We employed an NVIDIA RTX A6000 GPU with 48 GB memory for training and image generation. Two-point correlation functions for each fluid phase and cross-correlation functions between phases 1 and 2 guided the selection of a 64^3 voxel training size. This yielded 13,294 non-overlapping subvolumes. Our network, comprising 27 layers (1 input, 12 downsampling, 1 middle, and 12 upsampling), incorporates additional 3D transformer layers at specific down and upsampling blocks (8, 9, 11, 12 in downsampling and 1, 2, 4, 5 in upsampling). These transformers enhance global processing while controlling memory use.

Training for 30 epochs with a batch size of 8 took about 34 min per epoch. After training, generating 96^3 voxel images required only about 2 s each. While larger 3D images

demand more memory, this can be addressed using cloud computing or more powerful GPUs (An et al. 2017). As hardware improves and the model is further optimized, we anticipate handling image sizes exceeding the representative elementary volume (REV), enabling direct comparisons of our multiphase fluid reconstructions with current state-of-the-art methods.

While IPWGAN (Zhu et al. 2024) has proven effective at generating single-phase pore-space reconstructions, its performance in more complex multiphase scenarios remains unclear. To address this, we compared our DDIM-based approach with the state-of-the-art IPWGAN, known to produce pore-space images with geometric and flow properties closely matching training data. Using the same training dataset for both IPWGAN and DDIM ensures a fair comparison.

In addition to two-point and cross-correlation functions, we adopt a more comprehensive set of geometric and flow-related metrics. We use Minkowski functionals such as fluid-phase-specific surface area, curvature, and Euler characteristic to gain richer insights into pore topology and fluid phase morphology. Unlike two-point correlation functions, which primarily reflect basic spatial distributions, Minkowski functionals capture complex geometric features and connectivity patterns that directly influence flow.

We also evaluate global parameters including porosity, absolute permeability, and phase saturations (wetting and non-wetting). Porosity and permeability are critical for understanding fluid transport properties, while saturation quantifies fluid distribution within the pore network. Considering these global parameters provides a holistic view of the structural and hydraulic realism of the generated images, complementing and expanding upon the insights gained from two-point and cross-correlation analysis.

3 Results

The results of the generated images were evaluated from multiple perspectives, including direct visualization and comparisons of geometric and flow-related properties. Figures 2 and 3 illustrate the differences between images generated by DDIM, IPWGAN, and real samples. To perform the comparison, we generated 1000 synthetic images with the fully trained models and compared them against 1000 randomly selected experimental images of the same size. The analyses were conducted using commercial image analysis software and research codes (Raeini et al. 2012; Bijeljic et al. 2013).

First, as shown in Fig. 2, DDIM accurately reproduces the average geometric properties – such as curvature, interfacial area, and Euler characteristic – not only for the pore space but also for both fluid phases. Moreover, DDIM captures the entire distribution of these parameters, reflecting the statistical variability present in the training dataset. In contrast, while IPWGAN matches the average values, it fails to generate realistic parameter distributions. We found that IPWGAN struggles with accurately reproducing the multiphase fluid boundaries, adversely affecting geometric measures such as curvature and Euler characteristic, as evident in Fig. 2. Additionally, we find that IPWGAN's results exhibit a mode collapse (the generator only outputs similar samples, lacking diversity), resulting in a narrower range of parameter values than those observed in real images (Goodfellow et al. 2014). None of these issues arise in the DDIM-generated images.

Second, as shown in Fig. 3, DDIM-generated images closely resemble real samples, both in 2D slices and 3D visualizations. This resemblance clearly demonstrates the characteristic distribution of fluid phases: larger pores predominantly filled with

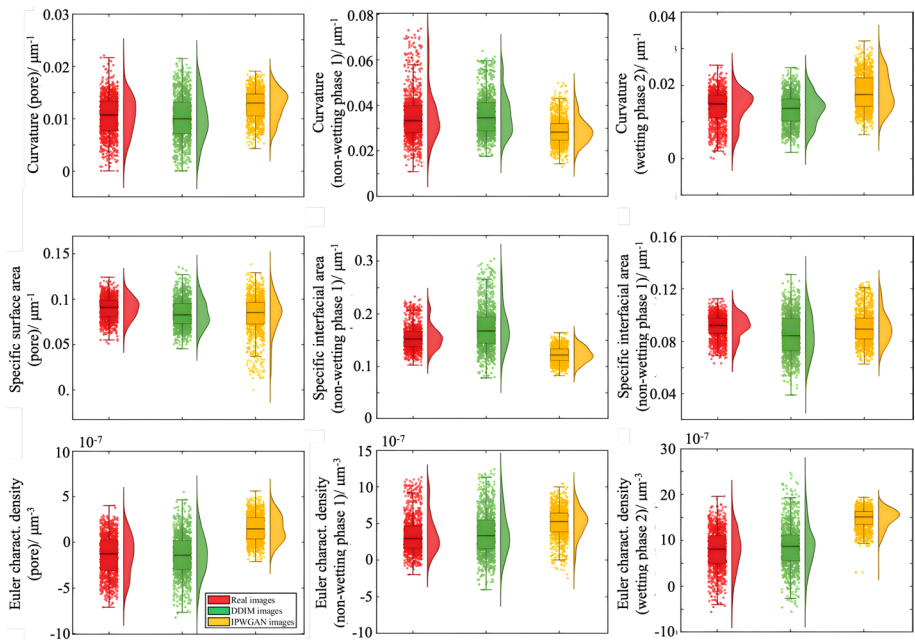


Fig. 2 Comparison of geometric properties. The plots show the distributions of curvature, Euler characteristic per unit volume, specific surface area, and specific interfacial area for the pore space (first column) and fluid phases (wetting phase 1 in the second column and non-wetting phase 2 in the third column). Comparisons are made between real, experimental images (red), DDIM predictions (green), and IPWGAN predictions (yellow). Each parameter is represented using a combination of box and violin plots. Box plots show quartiles, medians, and outliers; violin plots illustrate the distribution's shape. Statistics are based on 1000 randomly selected 96^3 samples from the entire image, with DDIM and IPWGAN generating an equal number of synthetic samples. DDIM-generated images exhibit parameter ranges, means, and other statistical characteristics closely resembling those of real images. In contrast, IPWGAN-generated results show deviations, especially in Euler characteristic and curvature, and display a reduced parameter range indicative of mode collapse. This demonstrates that DDIM-generated images are more physically realistic, reflecting the accuracy of diffusion models in image generation

the non-wetting phase (phase 2), and smaller pores primarily occupied by the wetting phase (phase 1). To quantitatively assess the quality of the generated images, we calculated parameters including porosity, absolute permeability using (Raeini et al. 2012; Bijeljic et al. 2013; Mostaghimi et al. 2013), curvature, specific surface area, and Euler characteristic.

Furthermore, the two-point correlation functions for the pore space, as well as the two-point cross-correlation functions between the non-wetting and wetting phases, closely match those of the real images. This includes the size distribution of non-wetting phase 2 clusters, as well as the distributions of porosity, permeability, and wetting phase 1 saturation. DDIM successfully replicates both voxel-level and volumetric features of the experimental 3D multiphase fluid images, along with their underlying physical characteristics.

Table S1 presents the range, median, and mean of all calculated parameters for the DDIM-generated and real images. The median and mean values from DDIM are very close to those of the real samples, underscoring the diffusion model's accuracy. Moreover, the range of parameter values in the DDIM-generated images aligns with that of the training dataset, indicating that the model can recall and reproduce a broad spectrum of

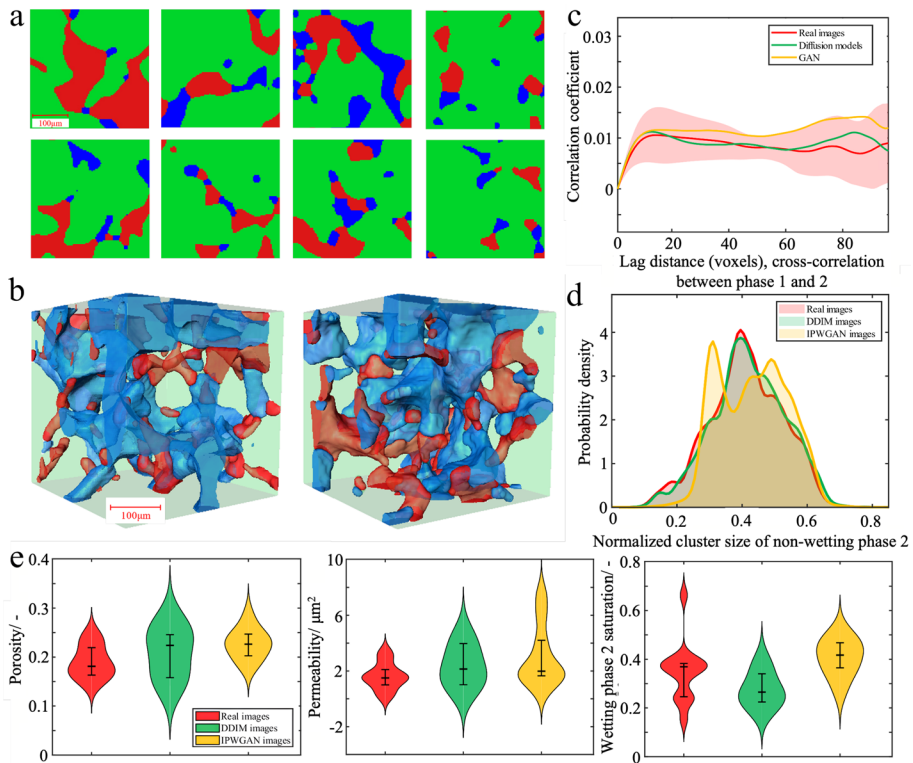


Fig. 3 Visual comparison of images, two-point correlation functions, cluster sizes, and permeability. **a** 2D slices: the first column shows four slices from real images, while the second column shows four slices from DDIM-generated images, revealing close visual resemblance. **b** 3D visualization: the left image is real, and the right is DDIM-generated, demonstrating strong 3D similarity. **c** Two-point cross-correlation function of phases 1 and 2 based on 1000 96^3 voxel images from real data, DDIM, and IPWGAN (as in Fig. 2). Red: real images; green: DDIM; yellow: IPWGAN. The shaded regions represent mean $\pm$ one standard deviation for the real data. **d** Normalized non-wetting phase 2 cluster size distribution computed using kernel density estimation. **e** Violin plots showing distributions of porosity, absolute permeability, and wetting phase 1 saturation. The box's central line represents the median, and the box the interquartile range, with whiskers extending to non-outlier data points. Violin width indicates data density

patterns. This diversity helps prevent the mode collapse typically associated with GAN-based methods.

4 Discussion

AI methods have successfully generated pore-space images, mitigating the long-standing trade-off between resolution and field of view. For example, previous work with IPWGAN showed that centimeter-scale porous media images can be reliably produced at micrometer resolution without distortion (Zhu et al. 2024). While the current diffusion model-based approach is still constrained by memory requirements and less mature optimization strategies, similar advancements may enable diffusion models to achieve the same balance in the future.

However, understanding fluid transport in porous media requires not only detailed pore-space representations but also information on fluid-phase distributions. Our diffusion model-based method directly reconstructs both the pore geometry and the multiphase fluid configuration. Although the current training is performed on 64^3 sub-volumes, initializing from a larger noise field allows generation of larger 3D images, potentially achieving a broader field of view without subdividing the domain. At present, generating 192^3 voxel images is the practical limit with a 48 GB GPU, but targeted optimizations, pruning, and knowledge distillation may help overcome this barrier (Rombach et al. 2022; Ma et al. 2024).

Recent innovations in diffusion models have reduced their memory requirements by factors of 10–1000, allowing them to match GANs in memory efficiency (Meng et al. 2023; Esser et al. 2024). Reduced memory usage also accelerates training. In addition, exploring the use of both dry and multiphase images as training data to generate realizations under various saturation conditions is a promising future direction. By tackling current memory constraints through model optimization and incorporating both dry and multiphase images into the training process, future research can further expand the applicability and efficiency of this approach. Leveraging machine learning to rapidly produce reliable images enables direct analysis of fluid characteristics under diverse scenarios, complementing experimental and simulation-based approaches and reducing reliance on expensive, time-consuming measurements, or computations.

Our long-term vision is to integrate experimental imaging, AI-driven image generation, and numerical simulations into a unified workflow. Experimental 3D pore-scale images provide a baseline, while the AI model rapidly approximates fluid configurations with different pore structures, wettability conditions, and flow parameters (Yan et al. 2024). The speed and accuracy of AI models can guide the selection of experimental conditions and refine simulation inputs. By iteratively combining experimentation, AI-driven generation, and simulation, researchers can efficiently explore complex parameter spaces, deepen their understanding of multiphase flow. Moreover, this approach can swiftly produce large sets of multiphase fluid pore-scale images of various sizes, enabling rapid and accurate analyses of pore structure-flow interactions and related physical parameters, all without the prohibitive costs of traditional methods. Rather than replacing conventional techniques, this data-driven strategy complements them, especially when the underlying physics of experimental observations remains uncertain.

The speed and flexibility of the diffusion model-based method are particularly valuable for optimizing fluid storage and transport systems, such as underground carbon dioxide and hydrogen storage, and for designing efficient electrolyzers and fuel cells. By applying AI to multiphase flow simulations, we substantially cut computational costs, broaden research capabilities, and open new avenues for industrial and environmental applications. Ultimately, this approach promotes the effective integration of experimental data and simulation outcomes, advancing the intelligent, data-driven analysis of fluid processes in porous media.

5 Conclusions

Our results demonstrate:

- (1) We have successfully applied diffusion models to generate 3D images that include both pore space and fluid phases. In principle, this approach can produce multiphase fluid the pore-scale images of arbitrary size, effectively addressing the long-standing challenge of obtaining large field of view, high-resolution multiphase images.
- (2) Applying our method to the Bentheimer sandstone dataset shows that it can accurately capture key fluid volumes, distributions, and interfacial characteristics. The generated images closely match real samples in terms of visualization, two-point correlation functions, and fluid and pore parameters. This level of fidelity holds substantial importance for accurately understanding and modeling fluid flow in porous media, as well as advancing research in petrophysics.

6 Appendix. Supplementary information

Download : It has been attached to the main text and merged into the same pdf. Refer to papers previously published in TIPM journals. After the official version is accepted, you can download the SI through the link provided by the journal.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s11242-025-02158-4>.

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Author Contributions Linqi Zhu was contributed conceptualization, formal analysis, funding acquisition, investigation, methodology, software, validation, writing—original draft, and writing—review and editing. Branko Bijeljic was involved in supervision, data curation, and writing—review and editing. Martin Blunt was performed conceptualization, data curation, project administration, supervision, resources, and writing—review and editing.

Data Availability Contact: zhulq@idsse.ac.cn

Python machine learning library: PyTorch

The source codes are available for download at: <https://github.com/ImperialCollegeLondon/DDIM>.

The dataset in this paper can be accessed through figshare data (analysis): https://figshare.com/articles/dataset/DDIM_image_result/26394880.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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